

Supplementary material

Resident and transient microbial taxa in the gill microbiome of the mussel

Mytilus galloprovincialis

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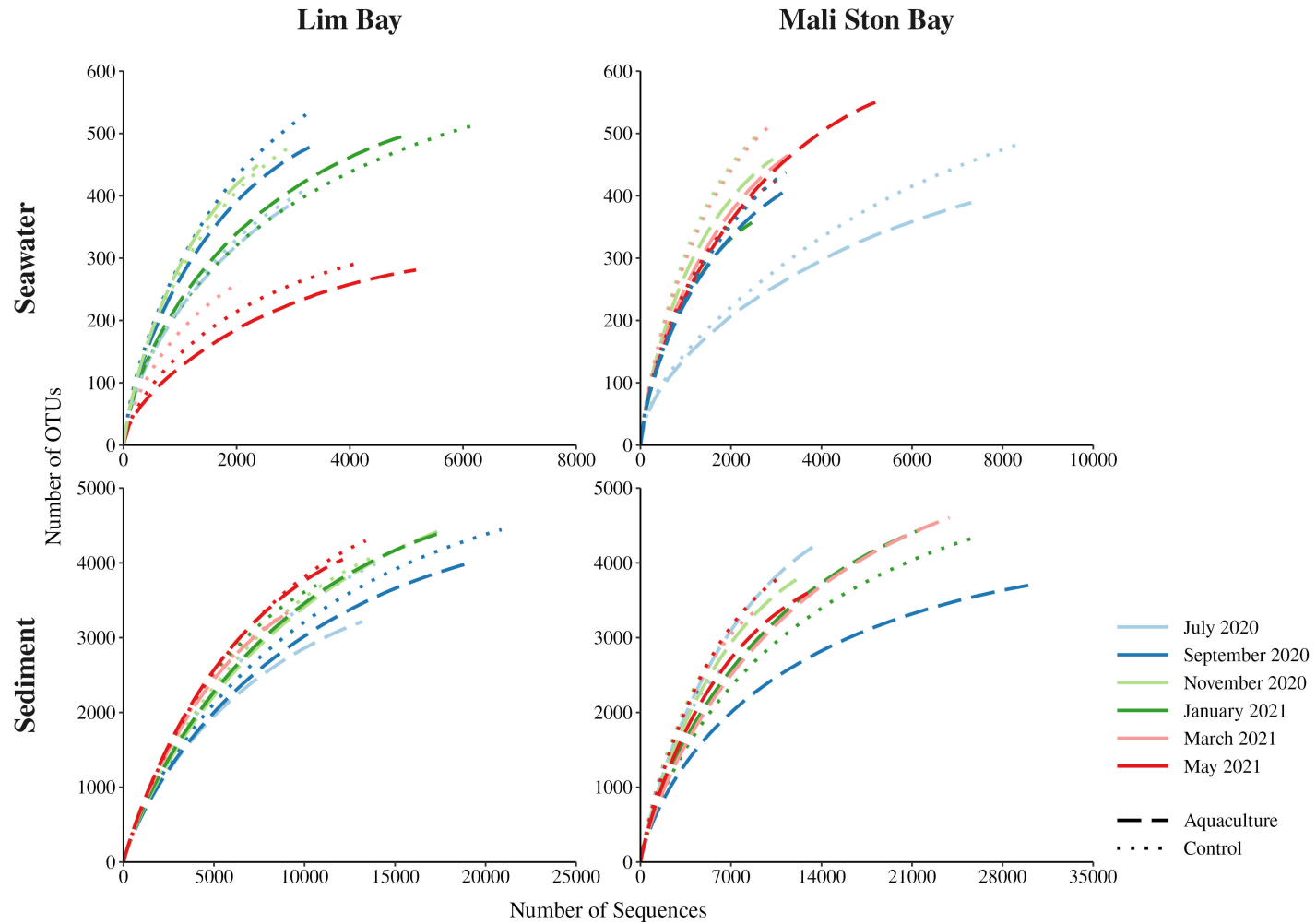
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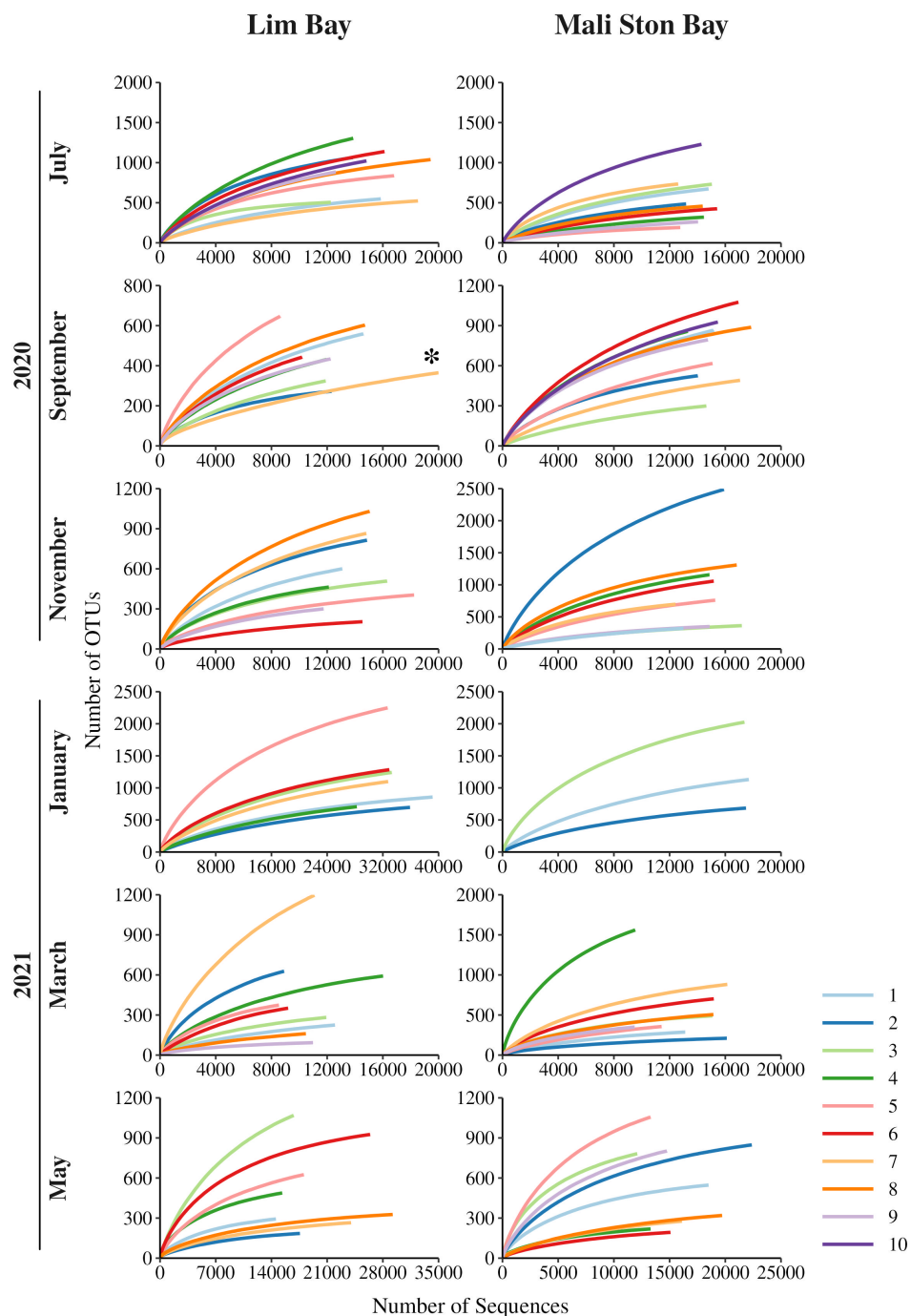
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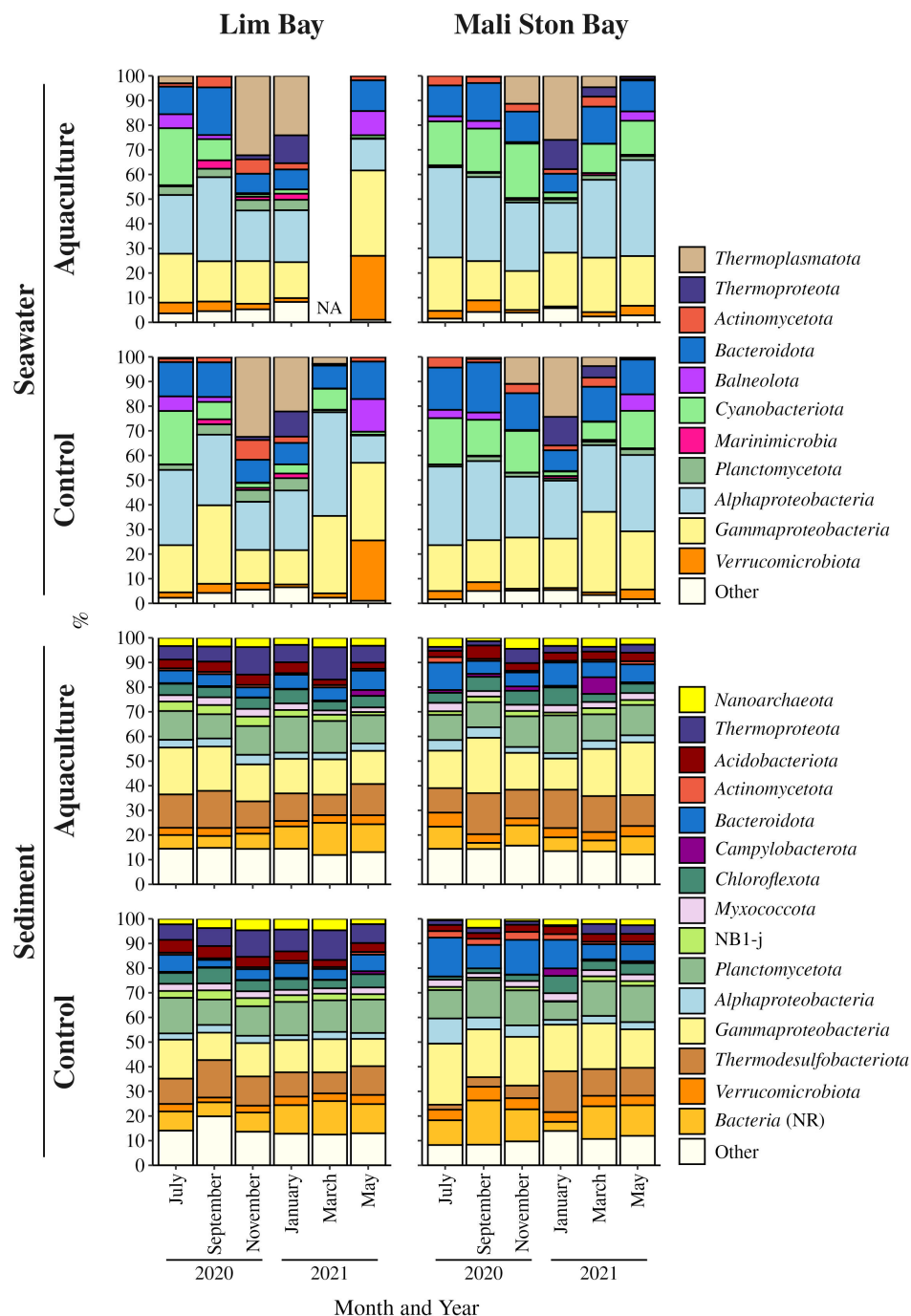
Supplementary figures



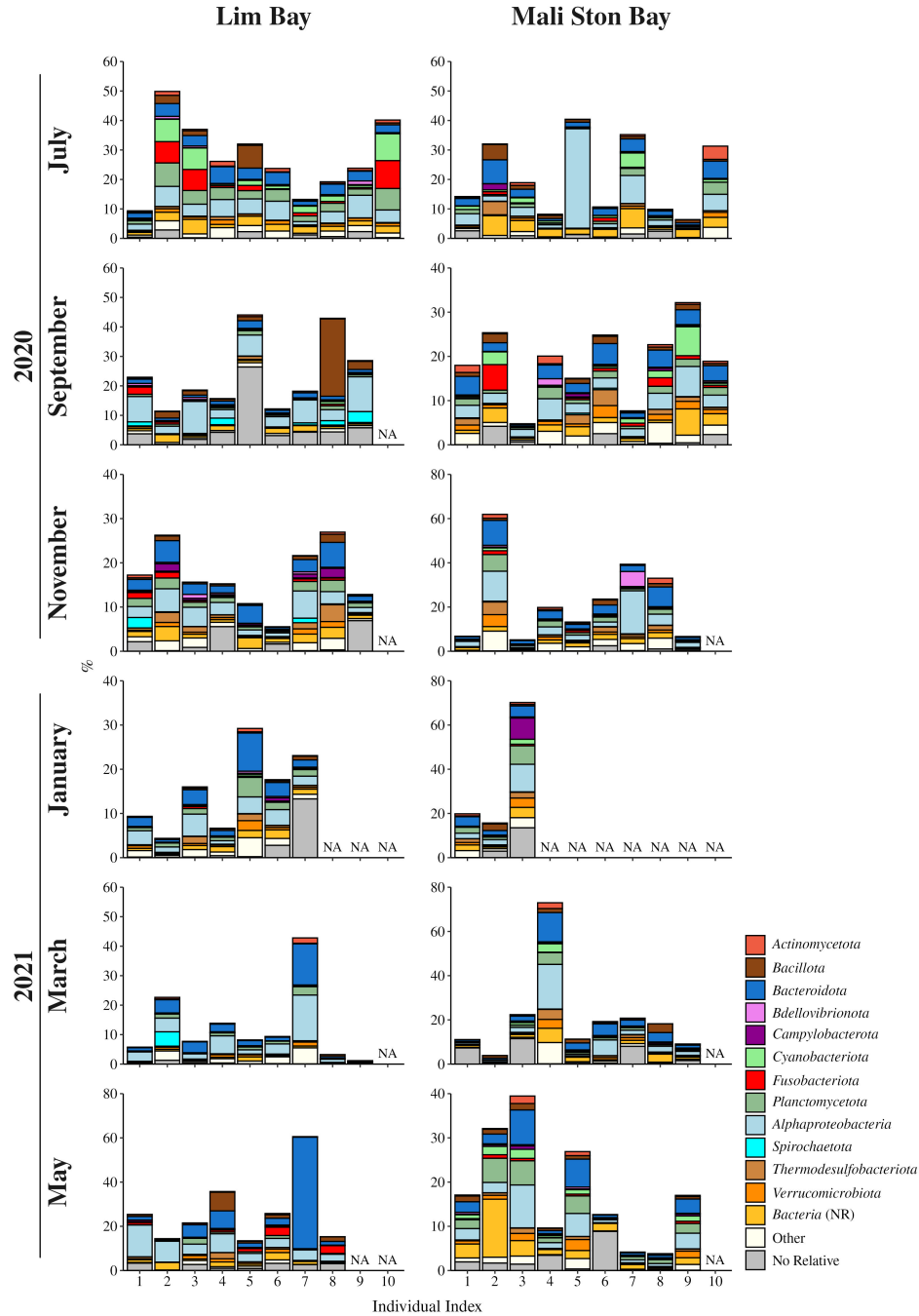
Supplementary Fig. S1 Rarefaction curves of seawater and sediment microbial communities sampled in Lim Bay and Mali Ston Bay from July 2020 to May 2021. In each bay, samples were collected from one aquaculture site and one control site



Supplementary Fig. S2 Rarefaction curves of microbial communities associated with the gills of the mussel *M. galloprovincialis* (gill microbiome). Mussels were collected in Lim Bay and Mali Ston Bay from July 2020 to May 2021. The numbers in the legend correspond to the index of each individual mussel (see Supplementary Table S2). The rarefaction curve marked with an asterisk (*) was truncated due to the high number of sequences obtained for this sample. This curve terminates at 60,611 sequences and 601 OTUs



Supplementary Fig. S3 Taxonomic classification and relative contribution of the most abundant ($\geq 3\%$) bacterial and archaeal sequences in the seawater and sediment of Lim Bay and Mali Ston Bay. In each bay, samples were collected from one aquaculture site and one control site between July 2020 and May 2021. “No Relative” indicates sequences that could not be classified into any taxon, while “NR” (No Relative) next to the name of a taxonomic group refers to sequences that could not be assigned to any known taxon within that group. “NA” (Not Available) indicates samples that were not successfully analysed



Supplementary Fig. S4 Taxonomic classification and relative contribution of the most abundant ($\geq 3\%$) bacterial and archaeal sequences from the gills of the mussel *M. galloprovincialis* (gill microbiome) excluding sequences classified as *Gammaproteobacteria*. From July 2020 to May 2021, 10 mussels were collected per sampling month in Lim Bay and Mali Ston Bay. Each bar (individual index) represents an individual mussel (see Supplementary Table S2). “No Relative” indicates sequences that could not be classified into any taxon, while “NR” (No Relative) next to the name of a taxonomic group refers to sequences that could not be assigned to any known taxon within that group. “NA” (Not Available) indicates samples that were not successfully analysed

Supplementary tables

Supplementary Table S1 Sample ID, sampling date, location, and site, and the no. of sequences and OTUs for each sample from the seawater and sediment environment. The no. of sequences and OTUs was calculated after excluding sequences without known relatives, as well as eukaryotic, mitochondrial, and chloroplast sequences

Sample ID	Environment	Date	Location	Site	No. of Sequences	No. of OTUs
1	Seawater	10 July 2020	Lim Bay	Aquaculture	3,037	390
2				Control	3,193	408
3	Seawater	7 July 2020	Mali Ston Bay	Aquaculture	7,309	389
4				Control	8,444	485
5	Seawater	19 September 2020	Lim Bay	Aquaculture	3,289	478
6				Control	3,225	530
7	Seawater	15 September 2020	Mali Ston Bay	Aquaculture	3,136	405
8				Control	3,212	438
9	Seawater	14 November 2020	Lim Bay	Aquaculture	2,376	449
10				Control	2,959	480
11	Seawater	10 November 2020	Mali Ston Bay	Aquaculture	2,931	459
12				Control	2,534	494
13	Seawater	15 January 2021	Lim Bay	Aquaculture	5,017	498
14				Control	6,296	516
15	Seawater	12 January 2021	Mali Ston Bay	Aquaculture	2,461	357
16				Control	2,772	413
18	Seawater	18 March 2021	Lim Bay	Control	2,062	262
19	Seawater	15 March 2021	Mali Ston Bay	Aquaculture	3,418	473
20				Control	2,952	518
21	Seawater	7 May 2021	Lim Bay	Aquaculture	5,169	281
22				Control	4,186	293
23	Seawater	3 May 2021	Mali Ston Bay	Aquaculture	5,188	550
24				Control	3,051	427
25	Sediment	10 July 2020	Lim Bay	Aquaculture	13,192	3,217
26				Control	14,292	4,027
27	Sediment	7 July 2020	Mali Ston Bay	Aquaculture	13,548	4,239
28				Control	7,194	2,722
29	Sediment	19 September 2020	Lim Bay	Aquaculture	19,217	4,006
30				Control	20,879	4,444
31	Sediment	15 September 2020	Mali Ston Bay	Aquaculture	30,251	3,708
32				Control	5,005	2,393
33	Sediment	14 November 2020	Lim Bay	Aquaculture	17,740	4,457
34				Control	13,887	4,090

Supplementary Table S1 Sample ID, sampling date, location, and site, and the no. of sequences and OTUs for each sample from the seawater and sediment environment. The no. of sequences and OTUs was calculated after excluding sequences without known relatives, as well as eukaryotic, mitochondrial, and chloroplast sequences (*continued*)

Sample ID	Environment	Date	Location	Site	No. of Sequences	No. of OTUs
35 36	Sediment	10 November 2020	Mali Ston Bay	Aquaculture Control	12,051 6,888	3,775 3,047
37 38	Sediment	15 January 2021	Lim Bay	Aquaculture Control	17,320 10,657	4,387 3,699
39 40	Sediment	12 January 2021	Mali Ston Bay	Aquaculture Control	21,574 26,060	4,439 4,352
41 42	Sediment	18 March 2021	Lim Bay	Aquaculture Control	9,264 9,305	3,359 3,367
43 44	Sediment	15 March 2021	Mali Ston Bay	Aquaculture Control	23,890 8,902	4,603 3,358
45 46	Sediment	7 May 2021	Lim Bay	Aquaculture Control	12,116 13,380	4,046 4,295
47 48	Sediment	3 May 2021	Mali Ston Bay	Aquaculture Control	12,924 10,778	3,595 3,808

Supplementary Table S2 Sample ID, sampling date and location, each individual ID and index, and the no. of sequences and OTUs for each gill sample of the mussel *M. galloprovincialis* (gill microbiome). The no. of sequences and OTUs was calculated after excluding sequences without known relatives, as well as eukaryotic, mitochondrial, and chloroplast sequences

Sample ID	Date	Location	Individual ID	Individual Index	No. of Sequences	No. of OTUs
49	10 July 2020	Lim Bay	1	1	15,853	548
50			2	2	13,847	1,064
51			3	3	12,266	505
52			4	4	13,872	1,304
53			5	5	16,808	837
54			11	6	16,120	1,139
55			7	7	18,523	523
56			8	8	19,428	1,038
57			9	9	12,622	884
58			10	10	14,824	1,022
59	7 July 2020	Mali Ston Bay	1	1	14,800	673
60			2	2	13,186	485
61			3	3	15,035	731
62			4	4	14,465	320
63			5	5	12,768	191
64			6	6	15,414	425
65			7	7	12,619	734
66			8	8	14,380	457
67			9	9	14,044	262
68			10	10	14,292	1,229
69	19 September 2020	Lim Bay	1	1	14,607	559
71			3	2	12,319	272
72			4	3	11,897	324
73			5	4	11,957	431
74			6	5	8,633	647
75			7	6	10,208	443
76			8	7	60,611	601
77			9	8	14,719	603
78			10	9	12,249	434
79	15 September 2020	Mali Ston Bay	1	1	15,175	866
80			2	2	14,029	525
81			3	3	14,641	299
82			4	4	13,308	858
83			5	5	15,088	617
84			6	6	16,943	1,076
85			7	7	17,061	491
86			8	8	17,856	888
87			9	9	14,753	794
88			10	10	15,466	927

Supplementary Table S2 Sample ID, sampling date and location, each individual ID and index, and the no. of sequences and OTUs for each gill sample of the mussel *M. galloprovincialis* (gill microbiome). The no. of sequences and OTUs was calculated after excluding sequences without known relatives, as well as eukaryotic, mitochondrial, and chloroplast sequences (*continued*)

Sample ID	Date	Location	Individual ID	Individual Index	No. of Sequences	No. of OTUs
89	14 November 2020	Lim Bay	1	1	13,090	600
90			2	2	14,861	814
91			3	3	16,309	508
92			4	4	12,120	463
93			5	5	18,244	404
95			7	6	14,538	204
96			8	7	14,815	865
97			9	8	15,054	1,030
98			10	9	11,746	300
99	10 November 2020	Mali Ston Bay	1	1	12,990	321
100			2	2	15,906	2,488
101			3	3	17,187	363
103			5	4	14,867	1,157
104			11	5	15,278	760
105			7	6	15,171	1,058
106			8	7	12,428	697
107			9	8	16,821	1,309
108			10	9	14,874	349
109	15 January 2021	Lim Bay	1	1	39,148	857
111			3	2	35,902	696
112			4	3	33,282	1,241
113			5	4	28,265	707
114			6	5	32,692	2,248
115			7	6	32,932	1,284
116			8	7	32,762	1,099
125	12 January 2021	Mali Ston Bay	7	1	17,689	1,131
127			9	2	17,488	685
128			10	3	17,378	2,025
129	18 March 2021	Lim Bay	11	1	12,564	225
130			12	2	8,908	628
131			3	3	11,939	282
132			4	4	16,016	592
133			5	5	8,540	374
135			7	6	9,198	351
136			8	7	11,089	1,197
137			9	8	10,472	160
138			10	9	10,986	93

Supplementary Table S2 Sample ID, sampling date and location, each individual ID and index, and the no. of sequences and OTUs for each gill sample of the mussel *M. galloprovincialis* (gill microbiome). The no. of sequences and OTUs was calculated after excluding sequences without known relatives, as well as eukaryotic, mitochondrial, and chloroplast sequences (*continued*)

Sample ID	Date	Location	Individual ID	Individual Index	No. of Sequences	No. of OTUs
139	15 March 2021	Mali Ston Bay	1	1	13,123	287
140			2	2	16,109	210
141			3	3	15,072	490
143			5	4	9,533	1,561
144			6	5	11,426	356
145			7	6	15,174	702
146			8	7	16,144	882
147			9	8	15,148	506
148			10	9	9,495	350
149	7 May 2021	Lim Bay	1	1	14,563	293
151			3	2	17,581	185
152			4	3	16,798	1,069
153			5	4	15,356	488
154			6	5	18,075	625
155			7	6	26,405	926
156			8	7	24,010	265
157			9	8	29,240	327
159	3 May 2021	Mali Ston Bay	1	1	18,499	547
160			2	2	22,381	848
161			3	3	12,097	782
162			4	4	13,295	219
163			5	5	13,295	1,056
164			6	6	15,082	193
165			7	7	16,106	277
167			9	8	19,715	320
168			10	9	14,766	802

Supplementary Table S3 The statistics of richness estimator and diversity index parameters of microbial communities from different environments and sites. Parameters were tested with the Kruskal-Wallis H test followed by pairwise comparisons using the Mann-Whitney U test. Bonferroni correction was applied to address the problem of multiple comparisons

Parameter	Kruskal-Wallis H test			Mann-Whitney U test	
	H	df	p	Comparisons Between Environments and Sites	p
Observed No. of OTUs	78.6	4	<0.0001	Seawater (Aquaculture)–Sediment (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Gills (Aquaculture)	<0.01
				Seawater (Aquaculture)–Seawater (Control)	1.00
				Seawater (Aquaculture)–Sediment (Control)	<0.0001
				Sediment (Aquaculture)–Gills (Aquaculture)	<0.0001
				Sediment (Aquaculture)–Seawater (Control)	<0.0001
				Sediment (Aquaculture)–Sediment (Control)	0.33
				Gills (Aquaculture)–Seawater (Control)	<0.001
				Gills (Aquaculture)–Sediment (Control)	<0.0001
				Seawater (Control)–Sediment (Control)	<0.0001
Chao1	60.1	4	<0.0001	Seawater (Aquaculture)–Sediment (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Gills (Aquaculture)	1.00
				Seawater (Aquaculture)–Seawater (Control)	1.00
				Seawater (Aquaculture)–Sediment (Control)	<0.0001
				Sediment (Aquaculture)–Gills (Aquaculture)	<0.0001
				Sediment (Aquaculture)–Seawater (Control)	<0.0001
				Sediment (Aquaculture)–Sediment (Control)	1.00
				Gills (Aquaculture)–Seawater (Control)	1.00
				Gills (Aquaculture)–Sediment (Control)	<0.0001
				Seawater (Control)–Sediment (Control)	<0.0001
ACE	60.1	4	<0.0001	Seawater (Aquaculture)–Sediment (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Gills (Aquaculture)	1.00
				Seawater (Aquaculture)–Seawater (Control)	1.00
				Seawater (Aquaculture)–Sediment (Control)	<0.0001
				Sediment (Aquaculture)–Gills (Aquaculture)	<0.0001
				Sediment (Aquaculture)–Seawater (Control)	<0.0001
				Sediment (Aquaculture)–Sediment (Control)	1.00
				Gills (Aquaculture)–Seawater (Control)	1.00
				Gills (Aquaculture)–Sediment (Control)	<0.0001
				Seawater (Control)–Sediment (Control)	<0.0001
Exponential Shannon	93.9	4	<0.0001	Seawater (Aquaculture)–Sediment (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Gills (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Seawater (Control)	1.00
				Seawater (Aquaculture)–Sediment (Control)	<0.0001
				Sediment (Aquaculture)–Gills (Aquaculture)	<0.0001
				Sediment (Aquaculture)–Seawater (Control)	<0.0001
				Sediment (Aquaculture)–Sediment (Control)	0.60
				Gills (Aquaculture)–Seawater (Control)	<0.0001
				Gills (Aquaculture)–Sediment (Control)	<0.0001
				Seawater (Control)–Sediment (Control)	<0.0001

Supplementary Table S3 The statistics of richness estimator and diversity index parameters of microbial communities from different environments and sites. Parameters were tested with the Kruskal-Wallis H test followed by pairwise comparisons using the Mann-Whitney U test. Bonferroni correction was applied to address the problem of multiple comparisons (*continued*)

Parameter	Kruskal-Wallis H test			Mann-Whitney U test	
	H	df	p	Comparisons Between Environments and Sites	p
Inverse Simpson	95.4	4	<0.0001	Seawater (Aquaculture)–Sediment (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Gills (Aquaculture)	<0.0001
				Seawater (Aquaculture)–Seawater (Control)	1.00
				Seawater (Aquaculture)–Sediment (Control)	<0.0001
				Sediment (Aquaculture)–Gills (Aquaculture)	<0.0001
				Sediment (Aquaculture)–Seawater (Control)	<0.0001
				Sediment (Aquaculture)–Sediment (Control)	1.00
				Gills (Aquaculture)–Seawater (Control)	<0.0001
				Gills (Aquaculture)–Sediment (Control)	<0.0001
				Seawater (Control)–Sediment (Control)	<0.0001